

MARKSMANSHIP COURSE MANUAL



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OVERWATCH TRAINING COMMAND

Kali Special Operations Legion, Headquarters

Foreword by the Commander

1. PURPOSE

Document KSOL-MK 3-1A (Marksmanship Course Manual) provides doctrine, principles, professional knowledge and procedures on Kali Special Operations Legion's rifle and DMR Marksmanship

2. SCOPE

This document covers the foremost and warfighting ethos and philosophy of Kali Special Operations Legion. This publication discusses and reflects the individual skills required for effective rifle and DMR (Designated Marksman Rifle) marksmanship.

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APPENDIX 'A'

Introduction

AA01. Marksmanship

Marksmanship is the ability to employ a firearm **accurately, consistently, and effectively** for the conditions and purpose the weapon is intended for.

Rifle marksmanship is the basic capacity to operate, and use a service rifle on its standard configuration accurately and competently by all rifleman.

However, **Designated Marksmanship** is a more specialized role with more advanced configuration, weapons, roles and overall equipments. Designated Marksmanship is primarily intended to engage at farther distance than rifle marksman but at closer distances than a sniper.

AA02. Know Your Firearm

Before starting, every personnel must be familiar with the standard issue rifle of KSOL regardless of their specialised roles/designation.

The standard issue rifle (as of writing this revision) is:

M27 (HK416 IAR) [in-game: RF416 A5]

To know more about the rifle, request an operator's manual for the firearm from the KSOL HQ.

Also, go to <https://ksol-hq.vercel.app/armoury.html> for more info on standard issue configuration.

APPENDIX 'B'

Weapon Handling

BA03. Purpose & Scope

Weapon handling is a consistent and standardized way for a combatant to effectively, and safely handle his weapon. Proper handling of a weapon ensures safety of the combatant and his fellow buddies. Weapon handling procedure applies at all levels of training or combat operations.

BA04. Safety Rules

Rule 1 – Treat every weapon as if it were loaded at all times.

Rule 2 – Never point your barrel towards anything you DO NOT intend to destroy.

Rule 3 – Keep your finger off the trigger (away from mouse button/ any button assigned to fire).

Rule 4 – Keep your weapon safety to SAFE until you intend to fire. (Press and hold your safety change key to put your weapon in safe. Usually, 'V' on keyboard. Press V again to undo it.)

BA05. Fire & Cease Fire

On command of "Fire", aim the rifle, put your safety off and fire.

On command of "Cease Fire", put the safety on.

APPENDIX 'C'

Marksmanship Fundamentals

CB06. Aiming

CB06 A. Sight Alignment

Sight alignment is the relation between foresight know/tip, rear sight and the firer's eye. This relationship is the most critical to fire a weapon accurately as intended.



CB06 B. Sight Picture

Sight picture is the placement of the foresight to the correct target position while maintaining proper sight alignment. Correct sight alignment with improper sight placement on the target will cause the projectile to impact wrong target.

CB07. Trigger Control

When firing in burst/full-automatic, a firer must have pressure control and panic control over his trigger (fire key) to avoid misfire, ammunition conservation and overall safety of the firer.

CB08. Follow Through & Recovery

Follow through is the continued application of the fundamentals until the round (projectile) has exited the barrel to avoid altering the impact/course of projectile. This includes keeping the weapon rested or steady as much as possible until the round exists the barrel.

Recovery is the action of aligning the sights back on the target or target direction immediately after the round exits the barrel to re-engage target(s) or for follow-up shots.

APPENDIX 'D'

Cover & Concealment

DC09. Overview

Cover protects a combatant from enemy fire, while **concealment** hides a combatant from enemy observation but may provide little or no protection.

Effective cover should ideally provide:

- Protection from direct and indirect fire
- Concealment from observation
- Support for the weapon
- A clear field of view/fire
- Safe, concealed access and exit

Common cover materials: packed earth, concrete, logs, sandbags, rubble, and other dense materials. No material should automatically be assumed to provide complete protection.

DC10. Types of Cover

a. Fighting Hole: Provides protection and a stable firing position. The position should be adapted to the combatant's height and available supports.

b. Rooftop: Use the roof structure for protection and support while minimizing exposure.

c. Window: Can provide concealment and support while allowing observation and engagement through the opening.

d. Vehicle: The vehicle's strongest structural areas can provide protection and support, but its level of protection varies greatly by vehicle. (Example: Engine blocks)

Key principle: A combatant should use available terrain and structures to **maximize protection and concealment while maintaining observation and an effective field of fire.**

APPENDIX 'E'

Zeroing

ED11. What is Zeroing?

Zero is the procedure of adjusting the weapon sights to cause the projectile to impact where the combatant wants it to. This must be done while compensating for weather effects, ballistics, projectile drop and range to the target.

ED12. Elements of Zeroing

ED12 A. Line of Sight

A straight line from the centre of the firer's eyes, through the rear sight and passing through the foresight knob/tip pointing towards the target.

ED12 B. Line of Fire

The path the projectile travels after exiting from the barrel to the point of impact is called the line of fire.

ED12 C. Centerline of a Bore

A straight line from the center at the camber end of the barrel, proceeding out of the muzzle, and continuing indefinitely.

ED12 D. Trajectory

Due to mass of projectile, wind, and physics, a projectile travels in an arc or curved flight path. Trajectory is the path a projectile travels to the impact. As a projectile exists the barrel it travels on an upward path, intersecting the light of sight (since the sights are placed above the barrel). As the projectile travels farther, it starts to drop down towards the Earth, intersecting the line of sight again.

ED12 E. Range

Range is the known distance from the target to the rifle.

ED13. Battlesight Zero (BZO)

BZO is the sight adjustment required to engage any point-sized targets from 0-300 yards/meters under ideal weather conditions (no wind).

APPENDIX 'F'

Field Craft

** Appendix 'F' is an abridged Roman Hindi to English Translation of Indian National Cadet Corps' Personal handwritten notes from 'Obi' aka Vandam. Expect mistakes. Review anything you read under appendix 'F' individually.*

FE14. Judging Distance

To fire a weapon effectively, a firer has to judge distance. A combatant usually will not engage targets more than 100 yards/meters, but one should have the ability to judge distance upto ~1000 yards/meters so that one knows:

- When to open fire.
- Accurately indicate target to supporting arms and sub-unit.
- Report correct & precise information as an observer.
- Determine which best-fit weapon to use for engagement.

FE15. Six (6) Methods of Judging Distance

FE15 A. Unit Method

A combatant visualizes 100 yards on the ground, and estimates how many such units would fit between the combatant and the target. The limitation of this method is its accuracy depends on the terrain visibility.

FE15 B. Method of Appearance

Distance (in yards/meters)	How a target appears
200	Full man with all his parts.
250	A kneeling man is covered by foresight knob.
300	Face turns diminish.
400	Face can't be recognized, foresight knob covers a standing person. Man's outline can be visible.
500	Body over shoulders look taper.
600	Heads appear as point, and body below shoulders look taper. They look wedge without a head.

FE15 C. Key Range Method

Using a known object/location/position's distance to judge distance between a firer and his target is key range method.

For example:

If firer is at point A, a telephone tower at point B and target at point C. *Given the known distance of Point A from B is 100m.*

A_____100m_____B_____C

To find the distance from A to C, one can estimate how many units of AB will cover the distance AC.

FE15 D. Halving Method

In halving method as the name suggests, firer estimates distance halfway to the target and doubles it.

FE15 E. Bracketing Method

In bracketing Method, combatant estimates the minimum possible (closest) and maximum possible (farthest) distance and finds the average of both the estimations.

For example: If a combatant estimates that the target's closest distance or target is at least 300 yards away, and the target's farthest distance or target is at maximum 500 yards away. Then the following is calculated:

$$(300 + 500) / 2 \text{ (Number of estimations)} = 400.$$

Therefore, the target is approx. 400 yards out.

FE15 F. Section Average

Two or more combatants make estimations using methods stated above and then find average of the estimation.

For example: Combatant A estimates 300yards, B estimates 250yards, C estimates 400yards.... then the calculation is:

$$(300 + 250 + 400) / 3 = \sim 300 \text{ yards outs}$$

FE16. Engaging Multiple Targets

FE16 A. Target Prioritization

Target priority is based on factors such as proximity & distance, opportunity, and threat-level. NO two situations will be same. Therefore, it solely depends on the combatants instincts, conscience, and common sense.

FE16 B. Technique of Engagement

- 1) Engage the first target with two-rounds or TRP (three-round press).
- 2) The recoil from the weapon can be used to direct the recovery of the weapon to the next target. Combatant has to physically move the sights to the desired target.
- 3) The above stated steps are repeated until all targets & threats are engaged.

FE17. Engaging Moving Targets

FE17 A. Types of Moving Targets

There are two types of moving targets: Moving targets, and stop-and-go targets.

A moving target moves consistently and remain in the field of view of the firer. Whereas, stop-and-go targets appear and disappear during its movements and these targets show up only for a brief period of time.

FE17 B. Leads

When a firearm is fired directly at a moving target, the target is in constant motion during the flight of the projectile discharge, this will cause bullets to drop and impact behind the intended target. Therefore, a combatant fires the weapon in front of the target depending on range, speed, velocity, projectile weight, windage, etc. This is called taking a lead.

APPENDIX 'G'

Designated Marksmanship

GF18. Purpose

A designated marksman is a specialized rifleman whose primary responsibility is to engage accurately and effectively at targets at ranges beyond the capacity of a regular rifleman.

The designated marksman is an integral part of an element but is not to be treated as a sniper.

GF19. Role of a Designated Marksman

A designated marksman is expected to be capable of:

- Identifying & engaging at targets at extended range.
- Providing effective cover fire to the integrated element.
- Observing and reporting relevant information to the element.
- Transitioning between precision and long-range engagements to regular rifleman roles when required.

GF20. MRADs/Mil dots

MRAD or Mil is short for "Milliradian". Mil is a unit for angular measurement. Other equivalent units are Degree (deg), Radian (rad), and MOA.

Mil and rad are usually used usually in physics and math, but we also use it for measuring distance in LR shooting. Without diving into mathematical and military jargons, lets take few things in simple overview.

- **Radian:** Radian is a measurement of arc.
- **Mil/MRAD:** Milliradian;

$$1 \text{ mil} = 1/1000\text{th of radian (or distance)} = 1/60\text{th of degree}$$

- There are approximately 6 radians in a circle (360 deg).
- **360 deg** = 6283 mil. Therefore, there are 6283 mil in a circle.

Demonstration(s):

*If a target is 1000 yards away then 1mil = 1/1000th of the distance.
Therefore, for 1000 yards target, 1mil = 1 yard.*

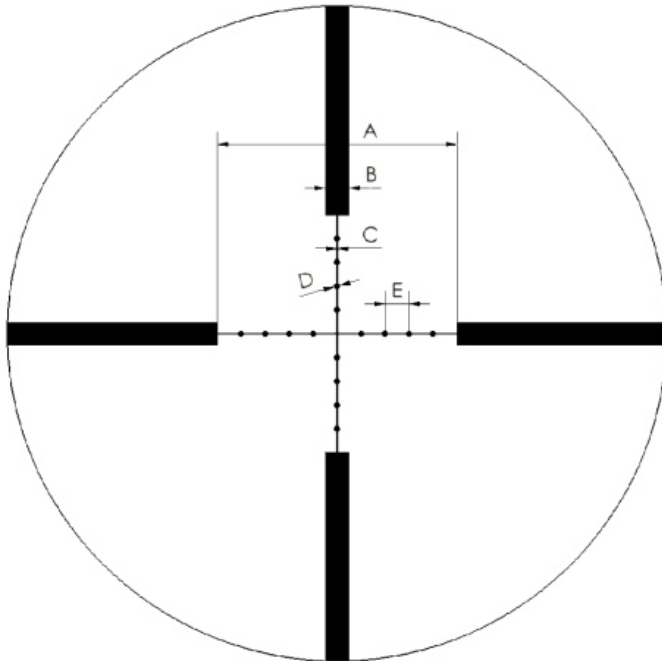
*If a target is 100 meters away (i.e.10,000cm) then 1mil = 1/1000th of the distance.
Therefore, for 100m/10,000cm target, 1mil = 10cm.*

GF21. MRAD/Mil Formula

$$(\text{Target size in m}) \times 1000 / (\text{Target size in mil}) = \text{Distance to target}$$

**m = meters; */ = division; *mil = milliradian*

****According to individual calculation and review (by me 'Obi' aka Vandam) the characters (including NPCs and players) have height of 0.3 meters or 30cm. This will help you to estimate range by worrying only about target size in milliradian which can be found by using your optic reticle. IT IS SUBJECT TO CHANGE.***

GF22. Mil dots on a reticle

This is a diagram of Schmidt & Bender PM3 scope reticle. The dots in the diagram are the mildots but the gap between one mildot to the other is not equal to a mil (milliradian).

The distance/gap from the center of one mildot to the center of other mildot is equal to a mil.

“MEN APART, EVERY MAN AN EMPEROR”

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